Paper Presentation

Classifying drivers of global forest loss

Deforestation versus temporary forest loss worldwide, 2001–2015

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RESEARCH

FOREST ECOLOGY

Classifying drivers of global forest loss

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Global maps of forest loss depict the scale and magnitude of forest disturbance, yet companies, governments, and nongovernmental organizations need to distinguish permanent conversion (i.e., deforestation) from temporary loss from forestry or wildfire. Using satellite imagery, we developed a forest loss classification model to determine a spatial attribution of forest disturbance to the dominant drivers of land cover and land use change over the period 2001 to 2015. Our results indicate that 27% of global forest loss can be attributed to deforestation through permanent land use change for commodity production. The remaining areas maintained the same land use over 15 years; in those areas, loss was attributed to forestry (26%), shifting agriculture (24%), and wildfire (23%). Despite corporate commitments, the rate of commodity-driven deforestation has not declined. To end deforestation, companies must eliminate 5 million hectares of conversion from supply chains each year.

Leaders of nearly 450 companies recently committed to zero deforestation in their supply chains by 2020 to meet consumer demand for deforestation-free products and to improve corporate social responsibility (*1*). Achieving these commitments requires transparency of complex supply chains for agricultural and forest products whose source locations are obscured by multiple aggregators and distributors (*2*). Large, multinational companies cannot determine the source of their supply beyond the location of their direct supplier, usually a distributor; this reduces the effectiveness of deforestation attribution and undermines a company's ability to take concrete action.

Companies, nongovernmental organizations, and governments are looking increasingly to data, maps, and tools to provide visibility on deforestation risk. Published maps of tree cover loss and gain derived from Landsat satellite observations (*3*) were a major step forward in consistent and transparent forest area change monitoring at a global scale. The launch of the online Global Forest Watch platform (*4*) extended the use and access of these data beyond the scientific community to include decision makers from governments, companies, and civil society organizations working to design and implement more effective forest policies.

However, the Hansen *et al.* dataset (*3*), updated annually on Global Forest Watch, does not distinguish permanent forest conversion associated with a change in land use [i.e., deforestation (*5*)] from other forms of forest disturbance that may be associated with subsequent regrowth (i.e., forestry, shifting cultivation, wildfire). This not only limits its utility for corporate decision-makers but also generates confusion when global forest cover change statistics derived from satellite imagery are compared directly against global land use change statistics as reported by governments in their national inventories (*6*). Deforestation involves the abrupt transition from land with trees to land without trees with no subsequent regrowth; loss of forest cover can also be associated with events such as wildfires, or with direct human-induced land use and land management practices such as clearcutting or selective logging, plantation forestry, smallholder agroforestry systems, or transitional subsistence farming due to shifting cultivation practices. As improvements in the spatial and temporal resolution of satellite imagery enable detection of smaller and more subtle changes to Earth's land surface relative to results from earlier monitoring efforts (*7*, *8*), more nuance is required in the attribution of global forest change dynamics.

Using high-resolution Google Earth imagery to visually classify nearly 5000 training sample cells, we developed a decision-tree model that predicts the most likely cause of forest disturbance at any 10 km × 10 km grid cell around the world since the year 2000 (*9*). Categories were assigned according to dominant disturbance type (Fig. 1), with each representing a different forest and land use dynamic: (i) commodity-driven deforestation, defined by the long-term, permanent conversion of forest and shrubland to a nonforest land use such as agriculture (including oil palm), mining, or energy infrastructure; (ii) shifting agriculture, defined as small- to medium-scale forest and shrubland conversion for agriculture that is later abandoned and followed by subsequent forest regrowth; (iii) forestry, defined as large-scale forestry operations occurring within managed forests and tree plantations with evidence of forest regrowth in subsequent years; (iv) wildfire, defined as large-scale forest loss resulting from the burning of forest vegetation with no visible human conversion or agricultural activity afterward; and (v) urbanization, defined as forest and shrubland

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Commodity-driven Deforestation | Shifting Agriculture | Forestry

Wildfire | Urbanization

Fig. 1. Representative examples of Google Earth imagery used to train the forest loss classification model. See (9) for more examples of training imagery.

Downloaded from https://www.science.org at University of Hong Kong on October 30, 2026

First page of the article (Science, 14 September 2018)

Background & Research Questions

Tree cover loss is not the same as deforestation

01 Pledges need location data

- Nearly **450 companies** pledged zero deforestation in supply chains by **2020**
- Sourcing passes through aggregators; firms rarely see beyond their **direct supplier**
- **Global Forest Watch** maps annual tree cover loss from Landsat (Hansen et al., 2013), but not its cause
- Loss maps compared directly with national land-use inventories create **confusion** (e.g. FAO Forest Resources Assessment)

Deforestation (paper's definition): an abrupt change from land with trees to land without trees, with no subsequent regrowth.

02 Loss maps show where, not why

	Permanent conversion	Temporary loss
Drivers	Commodity-driven deforestation; urbanization	Forestry; shifting agriculture; wildfire
After loss	Non-forest land use	Forest regrows

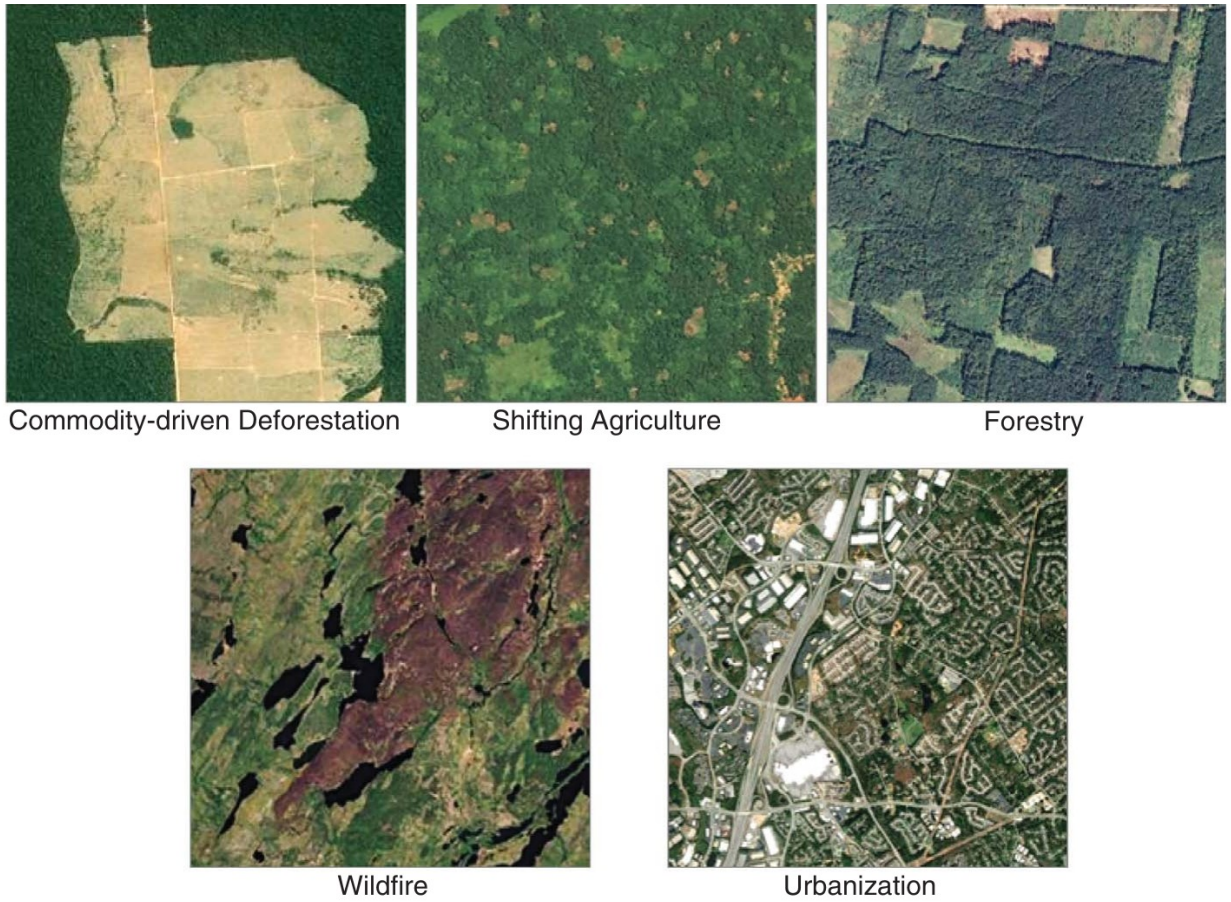
Research Questions

- *Which driver dominates forest loss in each 10 km cell, 2001–2015?*
- *How much of global loss is permanent deforestation?*
- *Has commodity-driven deforestation declined since 2001?*

Curtis, P. G. et al. (2018). Classifying drivers of global forest loss. Science, 361(6407), 1108–1111. Introduction, p. 1108.

Five Drivers of Forest Loss

Classes are defined by what happens after the trees are removed



Class	After the loss	Deforestation
Commodity-driven deforestation	Agriculture (incl. oil palm), mining, energy	**Yes**
Shifting agriculture	Farmed and abandoned; forest regrows	No
Forestry	Managed forest or plantation; regrows	No
Wildfire	Burned, no conversion; regrows slowly	No
Urbanization	Urban land	**Yes***

*Reported separately: the actors clearing land for cities differ from those clearing for commodities.

Curtis et al. (2018). Figure 1. Google Earth imagery used to train the model.

Curtis, P. G. et al. (2018). Classifying drivers of global forest loss. Science, 361(6407), 1108–1111. Fig. 1; pp. 1108–1109.

Classification & Validation

A decision tree trained on labelled imagery, checked by an independent sample

Input
Tree cover loss, Landsat 30 m, 2001–2015

→

Training
Nearly 5,000 cells labelled from Google Earth

→

Model
Decision tree: dominant driver per 10 km cell

→

Validation
1,565 random cells, labelled independently

01 Validation accuracy

Class / measure	Accuracy
Overall	89%
Commodity-driven deforestation	94%
Urbanization (smallest class)	**55%**
Forestry, North America (producer's)	96%
Forestry, Southeast Asia (producer's)	**78%**

02 Two estimates, one unit

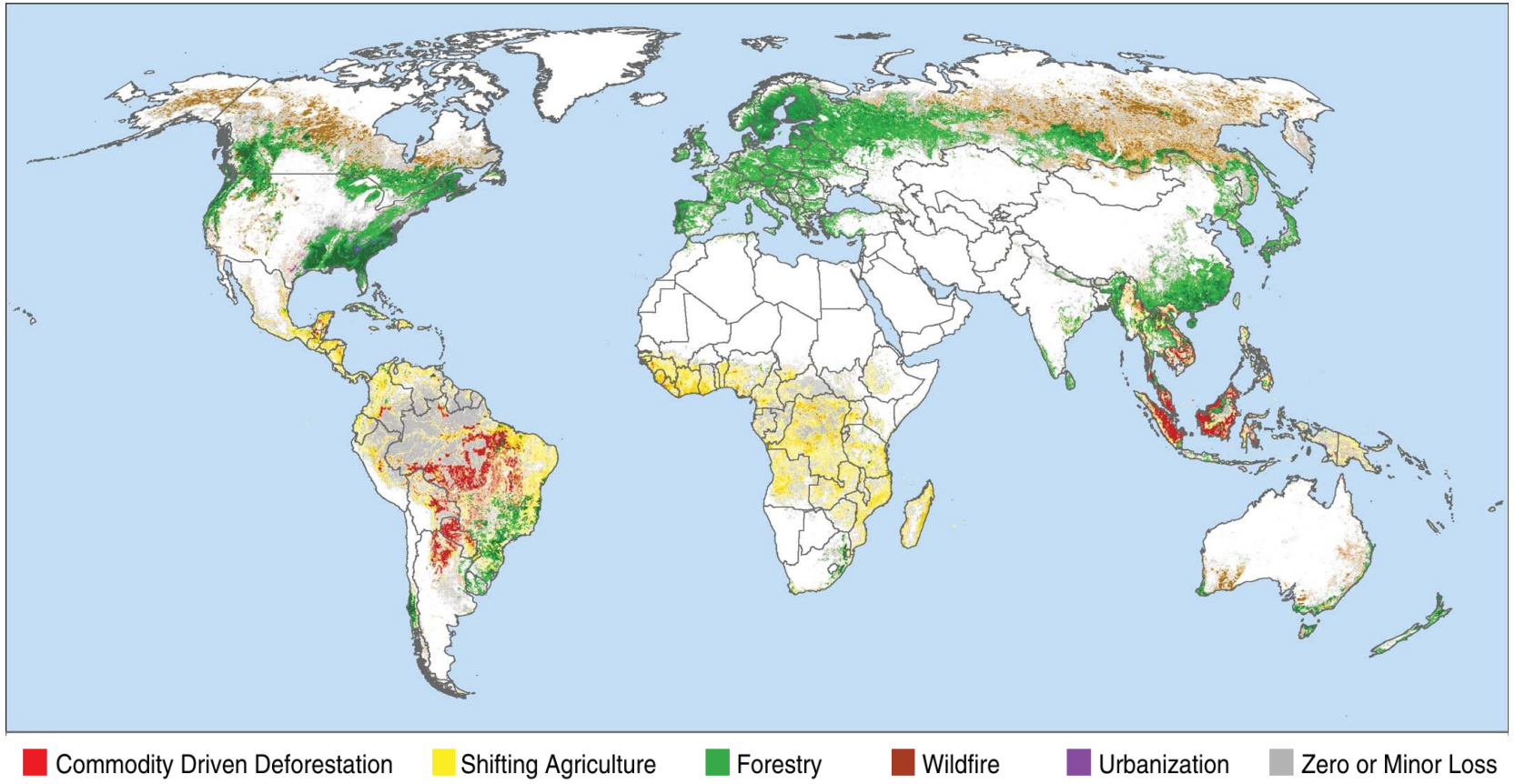
- **Map-based:** every cell with loss; locates sourcing regions
- **Sample-based:** 1,565 cells; shares ± 95% CI

Both estimates give each 10 km cell **one dominant driver**; the 95% CI therefore does not cover minority drivers inside a cell.

Curtis, P. G. et al. (2018). Classifying drivers of global forest loss. Science, 361(6407), 1108–1111. pp. 1108–1110.

Global Drivers, 2001–2015

About a quarter of forest loss is permanent deforestation



Curtis et al. (2018). Figure 2. Primary drivers of forest cover loss, 2001–2015.

Share of global loss

Sample-based, ± 95% CI (Table 1)

Driver	Share
Deforestation	**27 ± 5%**
Forestry	26 ± 4%
Shifting agriculture	24 ± 3%
Wildfire	23 ± 4%
Urbanization	0.6 ± 0.3%

- **Temperate & boreal:** forestry, wildfire
- **Tropics:** deforestation, shifting agriculture
- **≈ 3/4** of loss (≈ 230 Mha*) kept its land use

Curtis, P. G. et al. (2018). Classifying drivers of global forest loss. Science, 361(6407), 1108–1111. Fig. 2; Table 1. *Share × 314 Mha total loss.

Regional Contrasts

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Large loss totals do not mean large deforestation

Region	Map-based estimates: Hansen *et al.* (3)		Map-based estimates: Current study: Driver of tree cover loss					Sample-based estimates: Current study: Driver of tree cover loss				
	Tree cover loss (Mha, 2001–2015)	Tree cover loss (% of global total, 2001–2015)	Deforestation	Shifting agriculture	Forestry	Wildfire	Urbanization	Deforestation	Shifting agriculture	Forestry	Wildfire	Urbanization
North America	70	21%	1%	<1%	56%	40%	2%	2 ± 1%	1 ± 1%	48 ± 11%	48 ± 11%	1 ± 1%
Latin America	78	25%	56%	31%	13%	1%	<1%	64 ± 8%	24 ± 7%	9 ± 3%	<1 ± <1%	<1 ± <1%
Europe	15	5%	None	<1%	99%	1%	None	None	<1 ± <1%	95 ± 5%	5 ± 5%	None
Africa	39	13%	4%	92%	4%	<1%	<1%	2 ± 1%	93 ± 3%	4 ± 2%	<1 ± <1%	1 ± 2%
Russia/China/ South Asia	64	20%	<1%	<1%	41%	58%	<1%	2 ± 2%	1 ± 1%	38 ± 12%	59 ± 12%	<1 ± <1%
Southeast Asia	39	13%	78%	9%	13%	<1%	<1%	61 ± 13%	20 ± 10%	14 ± 6%	2 ± 6%	<1 ± <1%
Australia/ Oceania	10	3%	7%	10%	29%	53%	1%	8 ± 6%	10 ± 4%	19 ± 9%	62 ± 14%	1 ± <1%
Global	**314**	**100%**	**25%**	**21%**	**31%**	**22%**	**<1%**	**27 ± 5%**	**24 ± 3%**	**26 ± 4%**	**23 ± 4%**	**1 ± <1%**

Curtis et al. (2018). Table 1. Boxed: loss totals (Mha) and map-based deforestation shares discussed here.

- Latin America and Southeast Asia hold ≈ 74 of ≈ 77 Mha (≈ 96%) of mapped deforestation
- North America and Russia/China/S. Asia hold 43% of all loss but ≈ 1 Mha of deforestation: their loss is forestry and wildfire
- The paper cites a 2014 headline calling Canada the largest contributor to deforestation, a misreading of loss maps

Loss vs. deforestation*

Region	Loss	Defor.
Latin America	78	**43.7**
North America	70	0.7
Russia/China/S. Asia	64	<0.6
Africa	39	1.6
Southeast Asia	39	**30.4**
Europe	15	0
Australia/Oceania	10	0.7
All regions	**315**	**≈ 77**

**Mha. Deforestation = map-based deforestation share × regional loss (Table 1); regions sum to 315 Mha by rounding.*

Curtis, P. G. et al. (2018). Classifying drivers of global forest loss. Science, 361(6407), 1108–1111. Table 1.

Deforestation Did Not Decline

A flat global rate hides a moving frontier

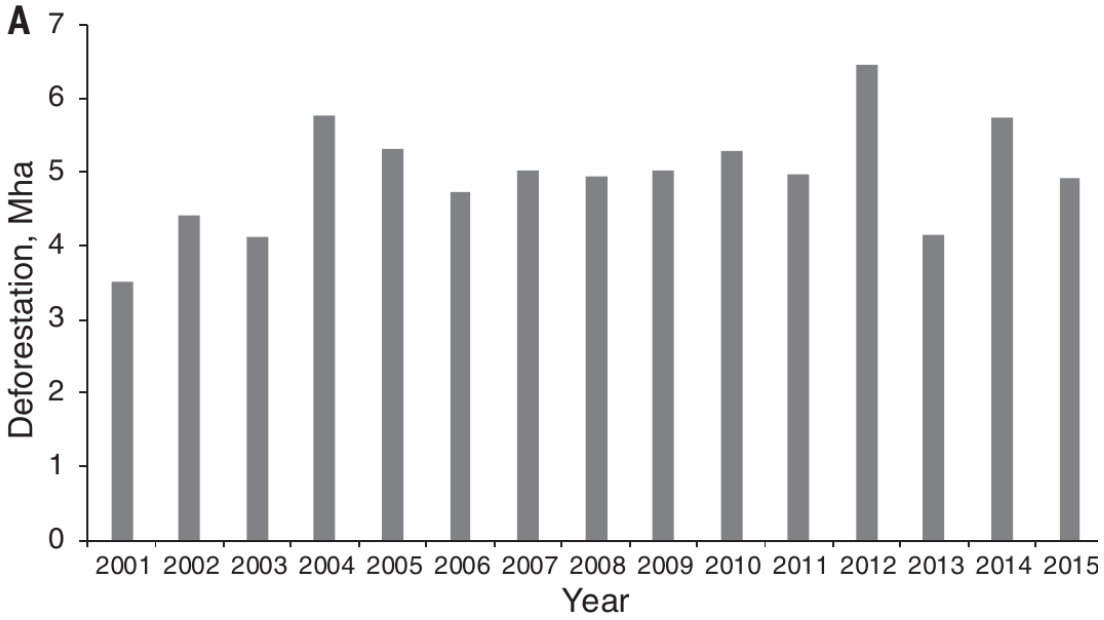


Figure 3A. Annual commodity-driven deforestation, worldwide.

- **≈ 5 Mha yr^{-1}** for 15 years, no downward trend
- 15 × 5 ≈ 75 Mha, matching 25–27% of 314 Mha*
- Ending it means removing ≈ 5 Mha of conversion from supply chains **every year**

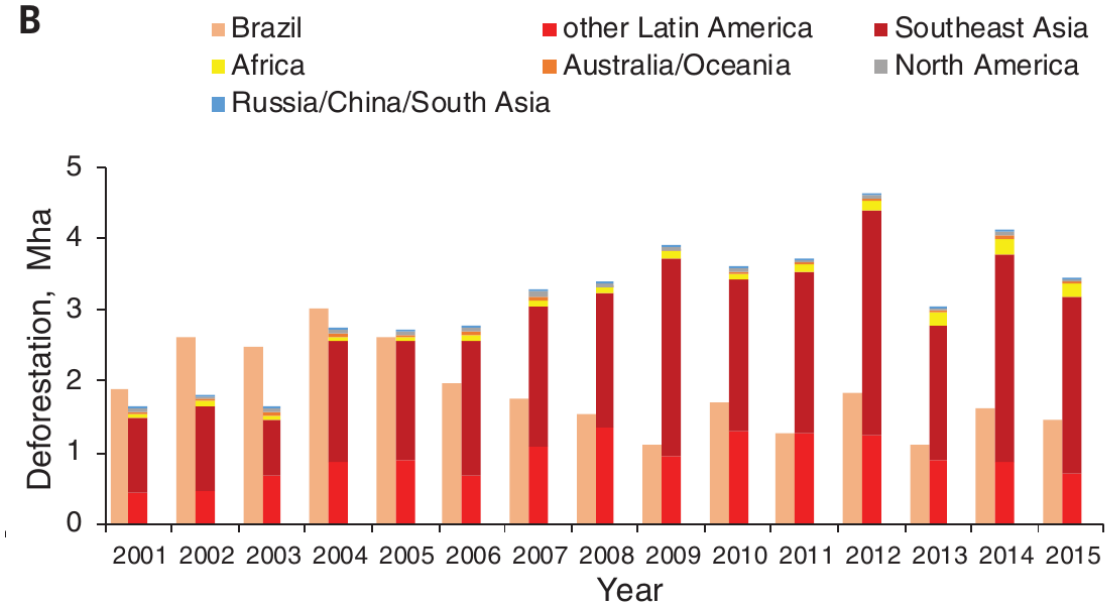


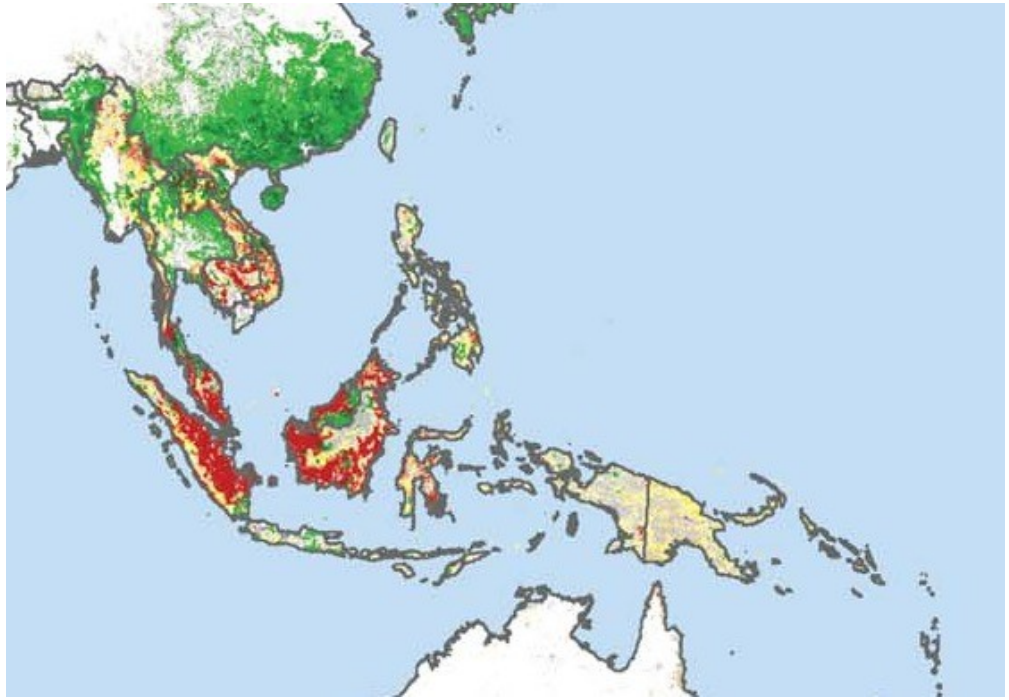
Figure 3B. Brazil and the rest of the world.

- Brazil falls; **Southeast Asia** and other Latin America rise
- Consistent with displacement; causality not tested
- A single-country success does not reduce the global total

Curtis, P. G. et al. (2018). Classifying drivers of global forest loss. Science, 361(6407), 1108–1111. Fig. 3; Table 1. *25–27% × 314 Mha ≈ 79–85 Mha.

Can the Map Be Trusted?

Map and sample estimates diverge where classes look alike



Curtis et al. (2018). Figure 2, Southeast Asia (cropped).

Red = commodity-driven deforestation.
SE Asia deforestation, map: 78%; sample: 61 ± 13%

01 Map outside the sample 95% CI

Region · driver	Map	Sample	Gap
SE Asia · deforestation	78%	61 ± 13%	**+17**
SE Asia · shifting agriculture	9%	20 ± 10%	**−11**
Australia/Oceania · forestry	29%	19 ± 9%	+10
Global · forestry	31%	26 ± 4%	+5

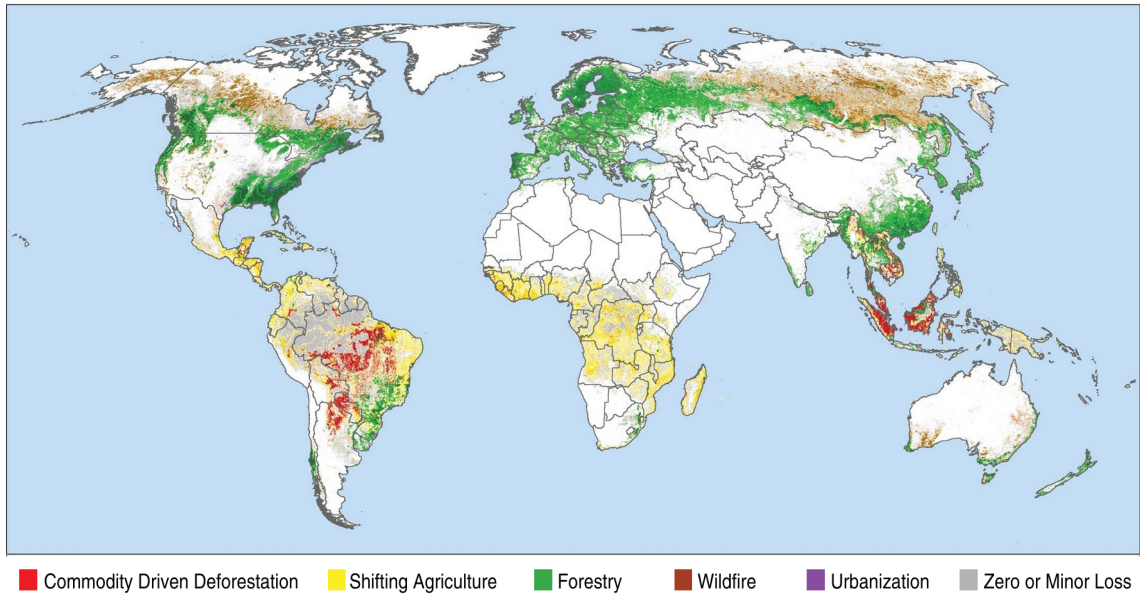
02 Design limits

- **Land-use definition:** forest → plantation counts as forestry
- **One driver per 10 km cell:** minority drivers disappear
- **Direct drivers only:** no link to markets or policy

Curtis, P. G. et al. (2018). Classifying drivers of global forest loss. Science, 361(6407), 1108–1111. Fig. 2; Table 1; pp. 1110–1111. Gap = map − sample (percentage points).

Conclusions

The map shows where forest loss is deforestation, and where it is not



Curtis et al. (2018). Figure 2. Primary drivers of forest cover loss, 2001–2015.

27 ± 5% of 2001–2015 forest loss was commodity-driven deforestation. Its rate stayed near **5 Mha yr^{-1}**: pledges are not being implemented at the pace needed for 2020. Ending deforestation means removing about 5 Mha of conversion from supply chains each year.

Implications by driver (authors' discussion)

Driver	Implication
Deforestation	Central & South America, Africa and SE Asia should be the focus of supply-chain efforts and REDD+ policies
Forestry	Not deforestation; exclude from zero-deforestation monitoring and use Fig. 2 as a wood-fibre sourcing map for certification
Shifting agriculture	Signals degradation and risk in smallholder food supply chains (slash-and-burn)
Wildfire	Nearly one-quarter of loss; not easily reduced through management

By separating deforestation from temporary loss, the study gives the more than 2 million Global Forest Watch users a way to avoid reading all tree cover loss as deforestation.

Curtis, P. G. et al. (2018). Classifying drivers of global forest loss. Science, 361(6407), 1108–1111. Abstract; pp. 1110–1111.